Mathematically, diversification benefits are measured by covariances or correlations. Denoting r_p as the portfolio return, the variance of the portfolio return is given by

$$\begin{aligned} \mathrm{var}(r_p) &= w_{US}^2 \mathrm{var}(r_{US}) + w_{JP}^2 \mathrm{var}(r_{JP}) + 2 w_{US} w_{JP} \mathrm{cov}(r_{US}, r_{JP}) \\ &= w_{US}^2 \mathrm{var}(r_{US}) + w_{JP}^2 \mathrm{var}(r_{JP}) + 2 w_{US} w_{JP} \rho_{US,JP} \sigma_{US} \sigma_{JP}, \end{aligned} \tag{3.1}$$

where r_{US} denotes U.S. returns, r_{JP} denotes Japanese returns, and w_{US} and w_{JP} are the portfolio weights held in the United States and Japan, respectively. The portfolio weights can be negative, but they sum to one, $w_{US} + w_{JP} = 1$, as the portfolio weights total 100%. (This constraint is called an *admissibility condition*.) The covariance, $\mathrm{cov}(r_{US}, r_{JP})$, in the first line of equation (3.1) can be equivalently expressed as the product of correlation between the United States and Japan ($\rho_{US,JP}$), and the volatilities of the United States and Japan (σ_{US} and σ_{JP}, respectively), $\mathrm{cov}(r_{US}, r_{JP}) = \rho_{US,JP}\sigma_{US}\sigma_{JP}$.

Large diversification benefits correspond to *low* correlations. Mathematically, the low correlation in equation (3.1) reduces the portfolio variance. Economically, the low correlation means that Japan is more likely to pay off when the United States does poorly and the insurance value of Japan increases. This allows the investor to lower her overall portfolio risk. The more Japan does not look like the United States, the greater the benefit of adding Japan to a portfolio of U.S. holdings. Mean-variance investors love adding investments that act differently from those that they currently hold. The more dissimilar, or the lower the correlation, the better.[7]

Figure 3.3 plots the United States and Japan mean-variance frontier for different correlation values. The solid line is the frontier with the 35.4% correlation in data. The dashed line is drawn with 0% correlation, and the dotted line drops the correlation to –50%. As the correlation decreases, the tip of the mean-variance frontier pushes to the left—the bullet becomes more pointed. The lower correlation allows the investor to reduce risk as Japan provides even more diversification.

2.3. G5 MEAN-VARIANCE FRONTIERS

In Figures 3.4 and 3.5 we add the United Kingdom, France, and Germany.

First consider Figure 3.4, which plots the mean-variance frontier for the G3: the United States, Japan, and the United Kingdom. The G3 frontier is shown in the solid line. For comparison, the old United States–Japan frontier is in the

[7] In this simple setting, there is only one period. In reality, correlations move over time and increase during bear markets. Ang and Bekaert (2002) show that international investments still offer significant diversification benefits under such circumstances. Christoffersen et al. (2013) report that correlations of international stock returns have increased over time but are still much lower in emerging markets than for developed ones. Investors also have access to frontier markets, which are the subset of the smallest, most illiquid, and least financially developed emerging markets.